Web Technologies (CSC336) - Lab Mid Term

Name: YOUR NAME
Registration No: FA24-BCS-118
N = 18 BP = 618px H = 126 K = 3 Sticky top = 88px Premium price > 1800

Screenshots rendered with Claude (headless Chromium) from the submitted files; product photos are shown as plain placeholder pictures.

Task 1 - CSS-only category filter

Screenshots

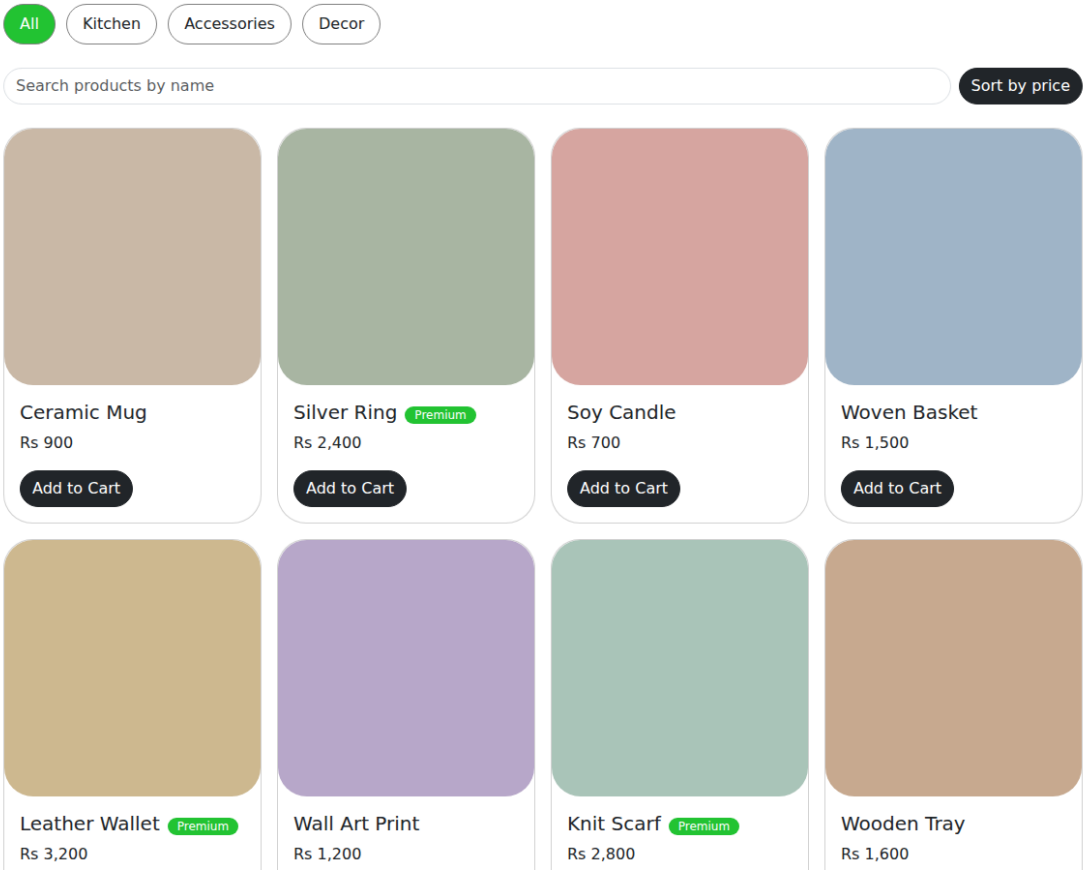
Task 1 at 375px - Kitchen selected



Task 1 at 618px (BP) - Accessories selected



Task 1 at 1280px - All selected



Task 1 - Code

index.html - radio buttons above the grid

```
<div class="filter mb-4">
  <input type="radio" name="cat" id="all" checked><label for="all">All</label>
  <input type="radio" name="cat" id="kitchen"><label for="kitchen">Kitchen</label>
  <input type="radio" name="cat" id="accessories"><label for="accessories">Accessories</label>
  <input type="radio" name="cat" id="decor"><label for="decor">Decor</label>
</div>
```

index.html - each product column has a category class

```
<div class="col kitchen">
  <div class="card">
    <a href="product.html"><img ... alt="Ceramic Mug"></a>
    <div class="card-body">
      <h5 class="card-title">Ceramic Mug</h5>
      <p class="price">Rs 900</p>
      <button class="btn btn-dark">Add to Cart</button>
    </div>
  </div>
</div>
<!-- Kitchen: Ceramic Mug, Wooden Tray
     Accessories: Silver Ring, Leather Wallet, Knit Scarf
     Decor: Wall Art Print, Woven Basket, Soy Candle -->
```

style.css

```
/* Task 1: category filter */
.filter input[type="radio"] { display: none; }
.filter label {
  padding: 8px 16px;
  border: 1px solid gray;
  border-radius: 30px;
  cursor: pointer;
  margin-right: 6px;
}
.filter input:checked + label {
  background: hsl(126, 70%, 45%);
  color: white;
}
body:has(#kitchen:checked)     #grid .col:not(.kitchen)     { display: none; }
body:has(#accessories:checked) #grid .col:not(.accessories) { display: none; }
body:has(#decor:checked)       #grid .col:not(.decor)       { display: none; }
```

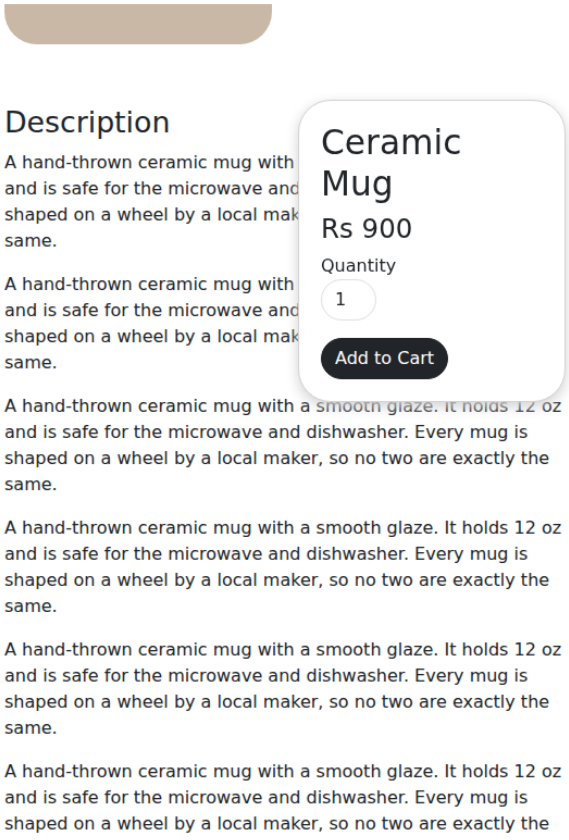
Task 2 - Product page layout with sticky details box

Screenshots

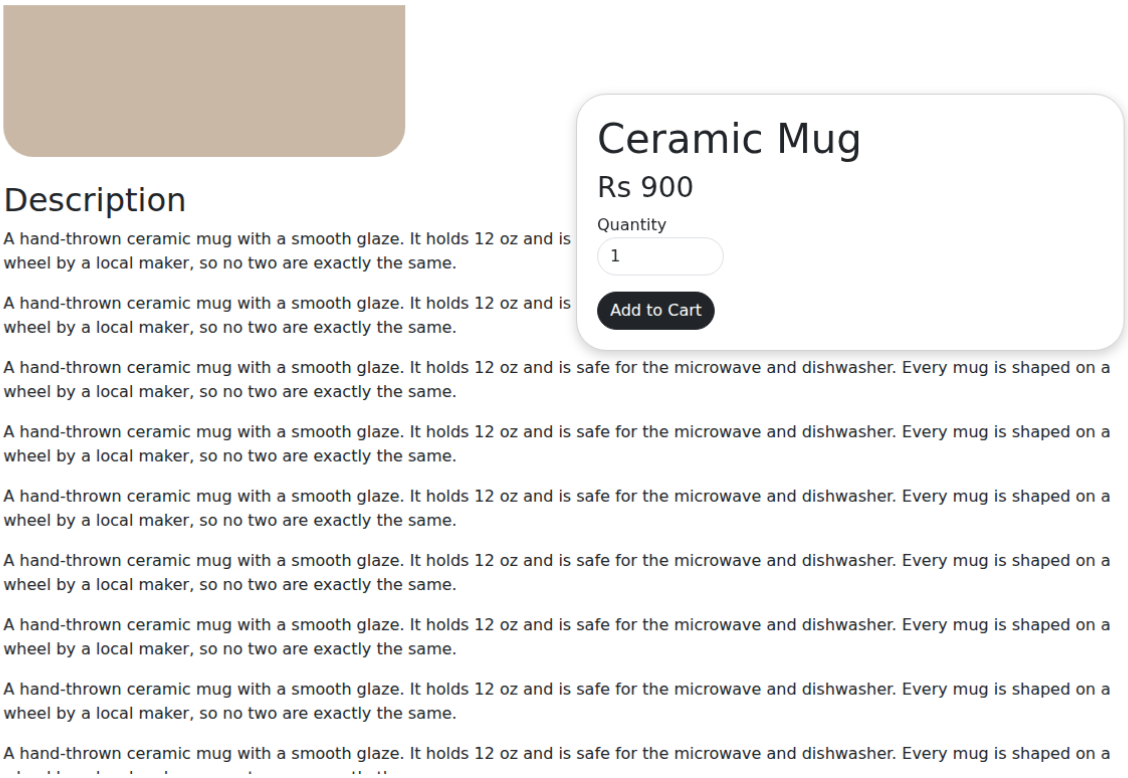
Task 2 at 375px - one column



Task 2 at 618px (BP) - side by side, scrolled, details box sticky



Task 2 at 1280px - side by side, scrolled, details box sticky



Task 2 - Code

product.html

```
<div class="product-page">

  <img src="..." class="img-fluid p-image" alt="Ceramic Mug">

  <div class="p-details">
    <h1>Ceramic Mug</h1>
    <h3>Rs 900</h3>
    <label for="q">Quantity</label>
    <input type="number" id="q" class="form-control w-25 mb-3" value="1" min="1">
    <button class="btn btn-dark">Add to Cart</button>
  </div>

  <div class="p-description">
    <h2>Description</h2>
    <p>A hand-thrown ceramic mug ... </p>   <!-- repeated 12 times -->
  </div>

</div>
```

style.css

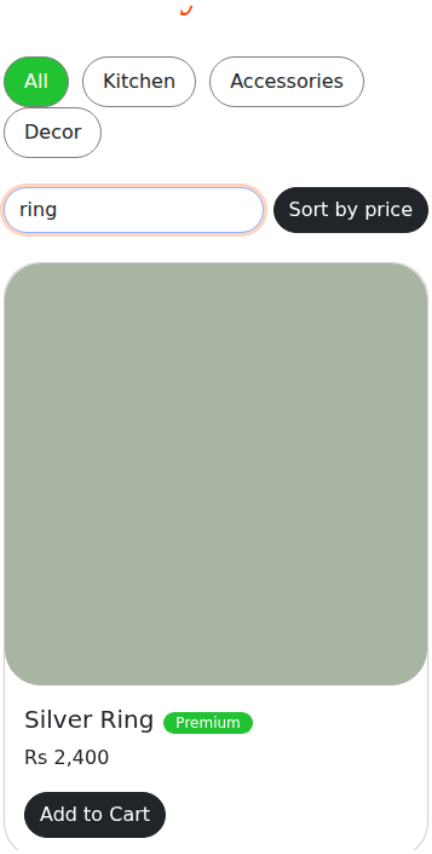
```
/* Task 2: product page grid (mobile first = one column) */
.product-page {
  display: grid;
  gap: 24px;
  grid-template-areas:
    "image"
    "details"
    "description";
}
.p-image       { grid-area: image; }
.p-details     { grid-area: details; }
.p-description { grid-area: description; }

@media (min-width: 618px) {
  .product-page {
    grid-template-columns: 1fr 1fr;
    grid-template-areas:
      "image details"
      "description description";
    align-items: start;
  }
  .p-details {
    position: sticky;
    top: 88px;
    z-index: 1;
    background: white;
    padding: 20px;
    border: 1px solid lightgray;
    border-radius: 30px;
    box-shadow: 0 4px 12px lightgray;
  }
}
```

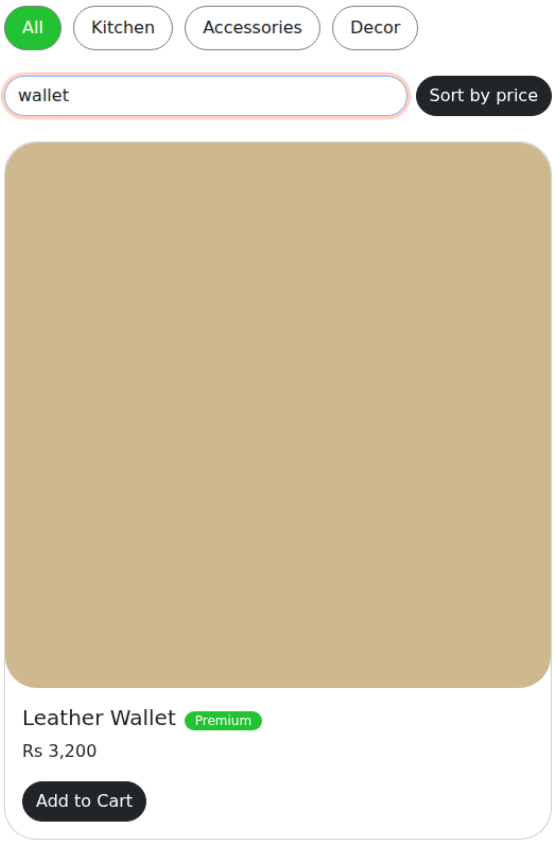
Task 3 - JavaScript live search and price sort

Screenshots

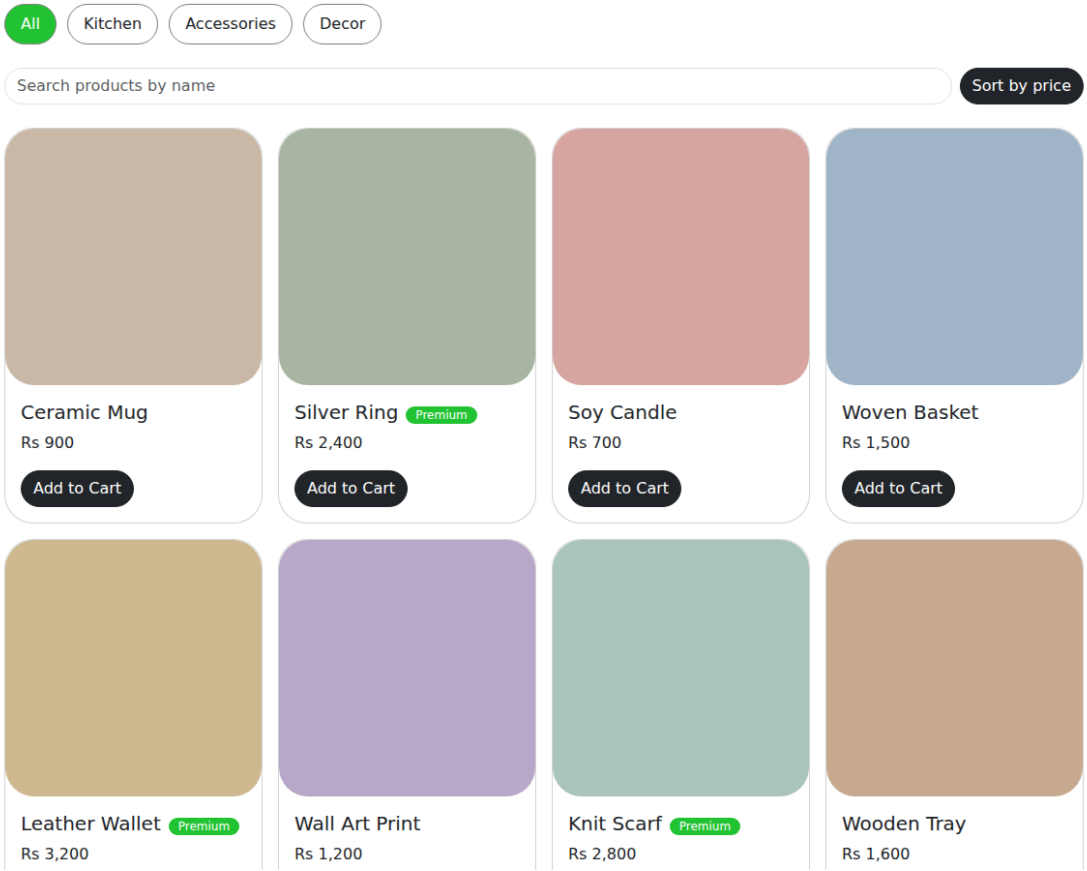
Task 3 - 375px - search for 'ring'



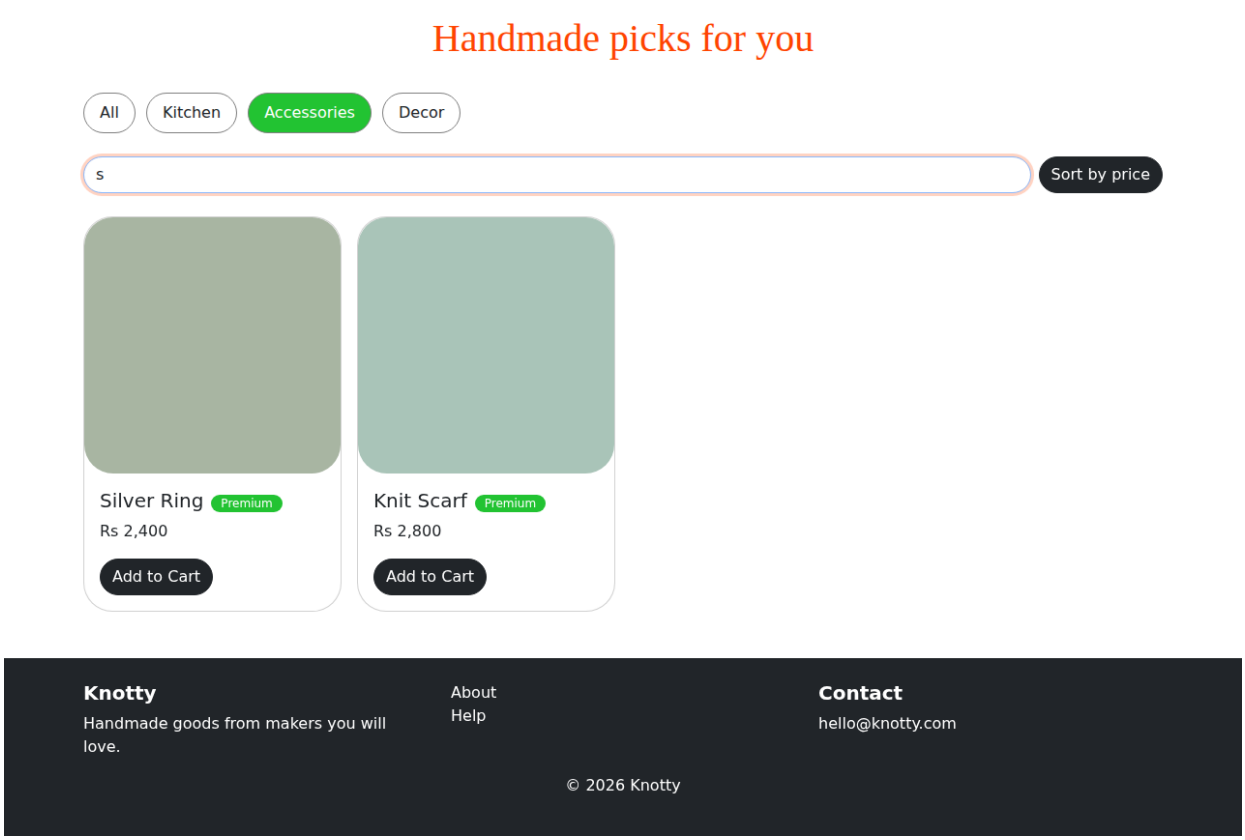
Task 3 - 618px (BP) - search for 'wallet'



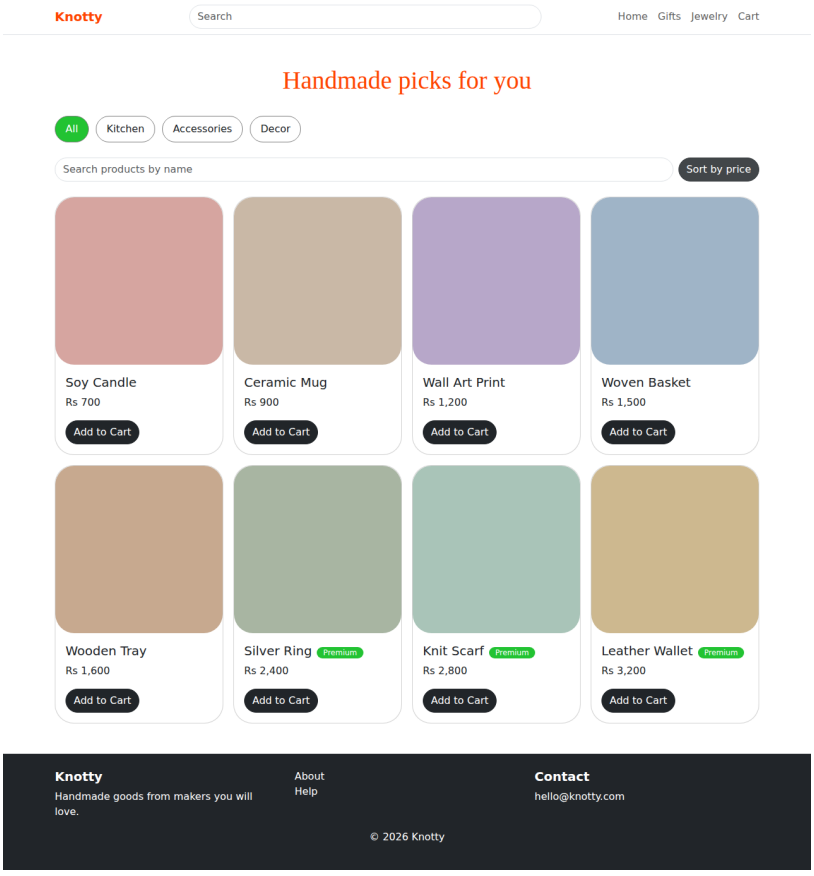
Task 3 - 1280px - Premium labels on prices above 1800



Task 3 - after search: Accessories category + 's' (both filters together)



Task 3 - after clicking Sort by price (lowest to highest)



Task 3 - Code

index.html

```
<!-- above the grid -->
<div class="d-flex gap-2 mb-4">
  <input type="text" id="search" class="form-control" placeholder="Search products by name">
  <button id="sortBtn" class="btn btn-dark text-nowrap">Sort by price</button>
</div>

<div id="grid" class="row row-cols-1 row-cols-md-2 row-cols-lg-4 g-3"> ... cards ... </div>

<!-- after the grid -->
<p id="no-results" class="text-center fs-4" style="display:none">No products found</p>

<!-- before </body> -->
<script src="script.js"></script>
```

style.css

```
/* Task 3 */
.hide { display: none; }
.premium {
  background: hsl(126, 70%, 45%);
  color: white;
  padding: 2px 10px;
  border-radius: 20px;
  font-size: 12px;
  margin-left: 8px;
}
```

script.js

```
const grid = document.getElementById("grid");
const cards = grid.querySelectorAll(".col");
const search = document.getElementById("search");
const message = document.getElementById("no-results");

// read the price number from a card ("Rs 8,900" -> 8900)
function getPrice(card) {
  const text = card.querySelector(".price").textContent;
  return parseFloat(text.replace(/[^0-9.]/g, ""));
}

// show the message when every card is hidden (by search OR by category)
function checkEmpty() {
  let visible = 0;
  cards.forEach(function (card) {
    if (getComputedStyle(card).display !== "none") {
      visible++;
    }
  });
  message.style.display = visible === 0 ? "block" : "none";
}

// live search by product name
search.addEventListener("input", function () {
  const text = search.value.toLowerCase();
  cards.forEach(function (card) {
    const name = card.querySelector(".card-title").textContent.toLowerCase();
    if (name.includes(text)) {
      card.classList.remove("hide");
    } else {
      card.classList.add("hide");
    }
  });
  checkEmpty();
});

// when a category radio changes, check again
document.querySelectorAll('input[name="cat"]').forEach(function (radio) {
  radio.addEventListener("change", checkEmpty);
});

// sort button: low to high, then high to low, and so on
let ascending = true;
```

```
document.getElementById("sortBtn").addEventListener("click", function () {
  const list = Array.from(grid.querySelectorAll(".col"));
  list.sort(function (a, b) {
    return ascending ? getPrice(a) - getPrice(b) : getPrice(b) - getPrice(a);
  });
  list.forEach(function (card) {
    grid.appendChild(card);
  });
  ascending = !ascending;
});

// Premium label on page load (price more than N x 100 = 1800)
cards.forEach(function (card) {
  if (getPrice(card) > 1800) {
    const tag = document.createElement("span");
    tag.textContent = "Premium";
    tag.className = "premium";
    card.querySelector(".card-title").appendChild(tag);
  }
});
```